Giacomo Negrinelli

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About

Senior Civil Hydraulic Engineer (Ing., MSc, BSc) and Project Manager with over 11 years of international experience in hydropower, dam safety, water resources, river engineering, and flood risk management. My expertise spans research, design, and construction, developed at leading organisations including Deltares (Netherlands), Frosio Next (Italy), Mott MacDonald (UK), and Gruner Stucky (Switzerland/Georgia). I am professionally qualified as Engineer in Italy (State Examination at University of Padua).

Work Experience

Project Manager / Engineer
Gruner Stucky SA, Renens (Switzerland)
Aug 2021 – Present

Within the Energy Business Area, I provide engineering consultancy services for the design and construction of dams and hydropower plants:

- Project management and design coordination.
- Hydropower planning and design.
- Design of hydraulic structures and hydraulic modelling.
- Dam safety and emergency preparedness.
- Geospatial analysis.

Project Engineer
Gruner Stucky Caucasus Ltd, Tbilisi (Georgia)
Oct 2021 – Dec 2023 · 2 yrs 3 mos

Office-based secondment to Gruner Stucky Caucasus Ltd, part of Gruner AG.

Civil Engineer
Mott MacDonald Ltd, Cambridge (United Kingdom)
Jan 2018 – Jul 2021 · 3 yrs 7 mos

Within the Dams, Rivers, and Geotechnics Group, I provided engineering consultancy services for the design and construction of dams, reservoirs and river hydraulic structures:

- Design and construction supervision of hydraulic structures.
- 1D/2D and CFD modelling of hydraulic structures.
- Project management and administration.
- Geospatial analysis.

Site Engineer
JN Bentley Ltd, Great Yarmouth (United Kingdom)
Oct 2019 – Mar 2020 · 6 mos

Site-based secondment for the Environment Agency Great Yarmouth Flood Defences Epoch 2 project:

- Setting-out, levelling and surveying.
- Verification of plans and drawings, and coordination of construction activities.
- Quality control and implementation of health and safety policies on site.

Hydraulic Engineer
Frosio Next S.r.l., Brescia (Italy)
Jan 2016 – Dec 2017 · 2 yrs

Within the BKW Engineering Network, I provided engineering consultancy services for the design, construction, and operation of hydropower schemes:

- Due diligence of hydropower plants.
- Hydrological analysis and hydropower potential assessment.
- 1D/2D hydraulic modelling.
- Geospatial analysis.

Language Proficiency

- **Italian** | *Native*
- **English** | *Full professional*
- **French** | *Limited working*

Countries of Work Experience

Australia · Belgium · Burkina Faso · Chile · Georgia · Italy · Laos · Mozambique · Philippines · Serbia · Slovenia · Switzerland · Tajikistan · Turkmenistan · UK · USA · Uzbekistan · Zambia · Zimbabwe.

Certifications

- **University of Padua**
  Section A Civil and Environmental Engineering Italian State Examination.
- **UK Construction Industry Training Board (CITB)**
  Site Management Safety Training Scheme (SMSTS), CSCS Academically Qualified Person Card.
- **BuildingSMART International**
  BIM Professional Certification Foundation.

Training

- Communication Pour Chefs de Projet at ISMAT Group SA, Lausanne (2026).
- Dam Risk Analysis Training at GNCOLD, Tbilisi (2023).
- BIM Methodology Training at Gruner AG, Renens (2023).
- 4x4 Off Road Training Course at Explore 4×4, Thetford (2021).
- Building Client Relationships and Collaboration at Mott MacDonald Academy, London (2019).
- FLOW-3D Workshop at EDF, Lyon (2019).
- Project Management Essentials at Mott MacDonald Academy, Croydon (2018).
- Presentation Skills at Mott MacDonald Academy, Cambridge (2018).

Teaching Assistant
University of Brescia, Brescia (Italy)
Feb 2017 – Sep 2017 · 8 mos

Teaching assistant for the course "Hydrology" (702998), Bachelor's Degree in Environmental and Land Engineering.

Hydraulic Engineer
Dizeta Ingegneria Studio Associato, Milano (Italy)
Nov 2015 – Dec 2015 · 2 mos

Freelance collaboration on a feasibility study for the reconstruction of a concrete dam on the Sauro River for water supply and irrigation, in the municipality of Agro di Aliano (Italy).

Graduate Intern
Deltares, Delft (Netherlands)
Mar 2015 – Aug 2015 · 6 mos

Research within the Dike Safety Department into flood embankment failure caused by backward erosion piping, involving physical and numerical modelling of seepage and erosion.

Graduate Intern
Consorzio di Bonifica Oglio Mella, Flero (Italy)
Jul 2014 – Sep 2014 · 3 mos

Internship with the public authority responsible for water resources management in the Oglio and Mella river basins:

- Streamflow measurement campaign.
- Technical assessment of hydropower projects.
- Hydraulic modelling of irrigation canals.

Education

Master's degree, Land and Environmental Engineering
University of Brescia, Brescia (Italy)
2013 – 2015

- **Grade:** 110/110 cum laude.
- **Main subjects:** Structural Design, Geotechnics, Hydraulic Structures, Environmental Hydraulics, Water and Wastewater Treatment, Geomatics, Urban planning, Risk Mitigation.
- **Thesis:** Investigation of the process of seepage and backward erosion piping under dikes in heterogeneous sands.

Erasmus+ Exchange Program, Civil Engineering
Technical University of Munich, München (Germany)
2014 – 2015

- **Grade:** 1.4 (Excellent).
- **Main subjects:** River Engineering, Hydro-morphology, Water Management, Alpine Hazards, Land Management and Policy, GIS.

Bachelor's degree, Civil Engineering
University of Brescia, Brescia (Italy)
2009 – 2013

- **Grade:** 110/110 cum laude.
- **Main subjects:** Mathematics, Physics, Chemistry, Structural Mechanics, Hydraulics, Hydrology, Geology, Surveying, Architecture, Building Technology, Economics.
- **Thesis:** An estimation procedure of synthetic design hydrographs for the design of flood control structures.

High School Diploma, Scientific Studies
Liceo Scientifico Statale "Nicolò Copernico", Brescia (Italy)
2004 – 2009

- **Grade:** 100/100.
- **Main subjects:** Scientific (Mathematics, Physics, Chemistry, Biology, Earth Science) and humanistic (Italian, Latin, History, Philosophy) studies.
- **Education level:** Lyceum (five-year university-preparatory school).

Resume

- HEC-RAS 2D Modelling Workshop at JBA Consulting, London (2018).
- Project Management course at ProjectGroup S.r.l., Brescia (2017).
- HEC-RAS 2D training course at H.S. S.r.l., Capannori (2016).
- Work Health and Safety training course at ConsulGroup S.r.l., Brescia (2016).

Publications

- Aykut, M., Micheloud, S., **Negrinelli, G.**, Bieri, M.P., Galdava, L., & Kopadze, T. (2024). *The Mtkvari hydropower project in Georgia: Dam safety upgrade of a project under execution*. In 2024 International Conference and Exhibition on Hydropower and Dams, Graz (Austria).
- Micheloud, S., **Negrinelli, G.**, Aykut, M., Galdava, L., & Kopadze, T. (2022). *A modern design approach at Mtkvari, Georgia*. Hydropower & Dams, 29(6).
- Cazzago, D., Papetti, L.L., & **Negrinelli, G.** (2016). *Design optimization of a very low head power plant - plenty of hydraulic common sense and a bit of CFD*. In 2016 International Conference and Exhibition on Hydropower and Dams, Montreux (Switzerland).
- De Lange, D.A., Van Beek, V.M., Schenkeveld, F.M., Van der Linden, T.I., & **Negrinelli, G.** (2016). *Preparation techniques for unconventional sand samples*. In 3rd European Conference on Physical Modelling in Geotechnics, Nantes (France).
- **Negrinelli, G.**, Van Beek, V.M., & Ranzi, R. (2016). *Experimental and numerical investigation of backward erosion piping in heterogeneous sands*. In 8th International Conference on Scour and Erosion, Oxford (UK).
- **Negrinelli, G.**, Van Beek, V.M., & Ranzi, R. (2016). *Small-scale piping experiments in heterogeneous sands*. In XXXV National Conference on Hydraulics and Hydraulic Structures, Bologna (Italy).
- Van Beek, V.M., **Negrinelli, G.**, Koelewijn, A.R., & Förster, U. (2015). *A coarse sand barrier as an effective piping measure*. Geotechniek, September 2015, 4-7.

Reference to Prior Work/Assignments

Rogun Hydropower Project Lot 2 Detailed Design

Date: 2024-Present
Location, Country: Rogun, Tajikistan
Client: Webuild S.p.A.
Project features: Detailed design under the EPC Lot 2 Dam Contract for the 335 m high earth-rockfill Rogun dam. Funded by the World Bank.
Position held: Team Lead Hydraulics
Activities: Technical and commercial coordination of multidisciplinary work packages related to reservoir sediment management, reservoir impoundment, diversion tunnel hydraulic analysis, dam break modelling, construction flood management, hydrological monitoring and forecasting, reservoir rim stability, dam safety risk assessment, and emergency preparedness planning.

Laos Irrigation Dam Safety Operation and Maintenance (LIDSOM)

Date: 2026
Location, Country: Vientiane Province, Laos
Client: Helvetas Swiss Intercooperation
Project features: Technical support to the Lao PDR Department of Irrigation (DOI) for completion of the feasibility study and institutional capacity strengthening for the rehabilitation of the 15 m high embankment Nam Houm and Nam Suang dams and the 24 m high embankment Nam Hin dam. Funded by the World Bank.
Position held: Technical Specialist
Activities: Feasibility design of the rehabilitation works for the spillways of Nam Houm and Nam Suang dams.

Framework Emergency Preparedness Plans for the Riparian Countries of the Vakhsh Cascade

Date: 2026
Location, Country: Turkmenistan and Uzbekistan
Client: Rogun Project Management Group (PMG)
Project features: Dam break inundation modelling, flood hazard mapping, consequence assessment, preliminary evacuation assessment, and preparation of framework emergency preparedness plans for Uzbekistan and Turkmenistan. The assignment also included recommendations for detailed hydraulic modelling, early-warning systems, public awareness and training, and priority early implementation measures. Funded by the World Bank.
Position held: Technical Specialist
Activities: Development of general recommendations on training and awareness raising for at-risk populations, identification of suitable early-warning systems for consideration by the riparian countries, and prioritisation of activities for early implementation.

Kozjak Pumped Storage Power Plant Basic Design Review and Tender Preparation

Date: 2026
Location, Country: Maribor, Slovenia
Client: Dravske elektrarne Maribor d.o.o. (DEM)
Project features: Consultancy services for the review of the updated basic design and preparation of tender documents for the 440 MW Kozjak pumped-storage hydropower plant.
Position held: Technical Specialist
Activities: Independent technical review, and participation in site inspection and workshops covering potential failure mode analysis, dam break inundation modelling, and consequence assessment in support of emergency preparedness plan development.

Sambuco Dam Heightening and Peccia Pumped Storage Hydropower Plant Upgrade

Date: 2026
Location, Country: Maggia Valley, Switzerland
Client: Officine Idroelettriche della Maggia SA (OFIMA)
Project features: Design of the heightening of the 130 m high arch-gravity Sambuco dam and of the upgrade of the 48 MW Peccia pumped-storage hydropower plant.
Position held: Technical Specialist

Activities: Hydraulic analysis of the preliminary study (SIA 31) for increasing the volume of the lower reservoir of the Peccia pumped-storage hydropower plant.

Agus-Pulangi Hydropower Complex Technical Due Diligence

Date: 2025-2026
Location, Country: Mindanao, Philippines
Client: Confidential
Project features: Technical due diligence of the existing generating assets and non-power components of the Agus 1, Agus 2, Agus 4, Agus 5, Agus 6, Agus 7, and Pulangi 4 storage hydropower plants, with a combined installed capacity of 1,000 MW.
Position held: Technical Specialist
Activities: Site inspections and assessments of the condition of hydropower facilities and associated structures, including powerhouses, dams, spillways, waterways, and penstocks.

Alimit Multi-Purpose Irrigation, Flood Control and Power Project Feasibility Study

Date: 2024-2026
Location, Country: Cordillera Administrative Region, Philippines
Client: SN Aboitiz Power Ifugao, Inc. (SNAP)
Project features: Value engineering and analysis study, and feasibility design of the 90 m high RCC gravity Alimit dam, the 120 MW Alimit storage hydropower plant, and the 20 MW Olilicon run-of-river hydropower plant.
Position held: Project Manager and Design Lead
Activities: Technical and commercial project management, including coordination and supervision of the technical design, management of administrative and financial matters, direction of technical and human resources, and quality assurance. Technical leadership of the scheme's hydrological study and civil hydraulic design, covering hydraulic modelling, spillway and outlet works, river diversion, and energy modelling.

Emergency Preparedness Plan for the Vakhsh Cascade

Date: 2024-2025
Location, Country: Vakhsh River, Tajikistan
Client: OJSC Barki Tojik
Project features: Preparation of the emergency preparedness plan for the Vakhsh Hydropower Cascade. Funded by the World Bank.
Position held: Technical Specialist
Activities: Development of guidelines for emergency actions, including identification and classification of dam safety emergencies and the corresponding response procedures for the 335 m high earth-rockfill Rogun dam (under construction) and the 300 m high Nurek earth-rockfill dam.

Oven Mountain Pumped Hydro Energy Storage Tender Design

Date: 2024
Location, Country: New South Wales, Australia
Client: Gamuda-Ferrovial JV
Project features: Tender design and drawings for the upper and lower concrete faced rockfill dams (CFRDs), reservoirs, and spillways of the 900 MW Oven Mountain pumped-storage hydropower plant.
Position held: Design Lead
Activities: Technical lead for the civil hydraulic design and tender drawings for the spillways, bottom outlets, and river diversion works associated with the 70 m high upper dam and 77 m high lower dam.

Wawa Pumped Storage Hydropower Plant Employer's Representative

Date: 2024
Location, Country: Calabarzon Region, Philippines
Client: Prime Infrastructure Capital Inc.
Project features: Design review, construction supervision and contractual support services for the 600 MW Wawa pumped-storage hydropower plant.
Position held: Technical Specialist
Activities: Hydraulic head loss analysis of the hydropower pressure system.

Licence Renewal of Resio Hydropower Plant

Date: 2024
Location, Country: Esine, Italy

Client: Italgen S.p.A.
Project features: Pre-feasibility study for the refurbishment of the 4 MW Resio run-of-river hydropower plant in support of participation in the operating licence renewal tender.
Position held: Technical Specialist
Activities: Hydrological analysis to assess water availability and identification of technical interventions to improve plant efficiency and minimise environmental impacts.

Enguri HPP Early-Warning System Feasibility Study

Date: 2023-2024
Location, Country: Samegrelo and Abkhazia, Georgia
Client: United Nations Development Programme (UNDP)
Project features: Feasibility study, tender design, technical specifications, and implementation planning for a flood forecasting and early-warning system for the 271.5 m high double-curvature arch Enguri dam.
Position held: Project Manager and Technical Specialist
Activities: Management of the project's administrative, financial, technical, human resource, and quality aspects. Technical lead for dam break and hydraulic modelling, flood hazard and risk assessment, and GIS mapping. Preparation of technical specifications and procurement requirements for the mass emergency notification system and hydro-meteorological monitoring equipment.

Namakhvani HPP Technical Due Diligence and Alternatives Study

Date: 2023-2024
Location, Country: Racha-Lechkhumi, Georgia
Client: Électricité de France S.A. (EDF)
Project features: Data collection, technical due diligence of the available input data, and development of the basis of design, alternative project configurations, and development plan for the 420 MW Namakhvani storage hydropower plant.
Position held: Project Manager
Activities: Coordination and supervision of the technical studies, management of the project's administrative and financial aspects, direction of technical and human resources, and oversight of compliance with quality requirements.

Enguri Pumped Storage Power Plant Feasibility Study

Date: 2023-2024
Location, Country: Samegrelo-Zemo Svaneti, Georgia
Client: Enguri Pumped Storage Power Plant LLC
Project features: Feasibility study, geological investigations, and environmental and social screening for a 570 MW pumped-storage hydropower plant located downstream of the Enguri dam.
Position held: Design Lead
Activities: Techno-economic assessment of alternative configurations for the dam, reservoir, and powerhouse. Preparation of the emergency preparedness plan framework for the scheme's 50 m high concrete-faced rockfill dam.

Qveshkhi HPP Project Review and Detailed Design

Date: 2023
Location, Country: Zemo Svaneti, Georgia
Client: Euroline LLC
Project features: Definition of the final project layout, detailed design of the civil works, and concept design of the electromechanical and hydromechanical equipment for the 5 MW Qveshkhi run-of-river hydropower plant.
Position held: Design Lead
Activities: Coordination of the multidisciplinary design team, site inspection, development of the concept design and final project layout, and preparation of energy and financial models.

Mtkvari HPP Owner's Engineer Services

Date: 2021-2024
Location, Country: Samtskhe-Javakheti, Georgia
Client: Mtkvari Hesi LLC
Project features: Owner's Engineer services for the 53 MW Mtkvari storage hydropower plant, including feasibility and detailed design of the main spillway for the scheme's 28 m high geomembrane-faced embankment dam.
Position held: Design Lead

Activities: Coordination of the multidisciplinary design team, civil hydraulic design of the main spillway and the small hydropower plant for environmental flow releases, 2D hydraulic modelling using HEC-RAS, and review of the civil hydraulic designs prepared by the Client's consultants.

Hydrological Model and Equipment Support for the Enguri Hydrology Initiative

Date: 2023
Location, Country: Zemo Svaneti, Georgia
Client: European Bank for Reconstruction and Development (EBRD)
Project features: Definition of the technical requirements for the procurement and installation of hydro-meteorological monitoring stations in the Enguri River basin.
Position held: Project Manager and Technical Specialist
Activities: Management of the project's technical, financial, and administrative aspects, conduct of site inspections, development of tender designs for the extension of the existing hydro-meteorological monitoring network in the Enguri River basin, and provision of technical support to the Client.

Study of Khudoni HPP Project Alternatives and Selection of the Final Layout

Date: 2023
Location, Country: Zemo Svaneti, Georgia
Client: Georgian Energy Development Fund (GEDF)
Project features: Technical, social, environmental, and economic assessment of design alternatives for the 700 MW Khudoni storage hydropower plant and identification of the preferred project configuration.
Position held: Technical Specialist
Activities: Energy, economic, and financial modelling, SWOT analysis, and development of the multicriteria decision matrix.

Enguri Hydrology Initiative

Date: 2021-2023
Location, Country: Zemo Svaneti, Georgia
Client: Georgian Energy Development Fund (GEDF)
Project features: Consultancy services to establish a river basin-wide initiative for hydro-meteorological data collection, hydrological modelling, and forecasting in the Enguri River basin. Funded by the European Bank for Reconstruction and Development (EBRD).
Position held: Technical Specialist
Activities: Provision of technical advisory services and capacity building in hydro-meteorological monitoring and measurement techniques. Preparation of guidelines for the planning, installation, operation, and maintenance of hydro-meteorological monitoring equipment.

Ouagadougou Water Supply Project

Date: 2022
Location, Country: Ouagadougou, Burkina Faso
Client: Office National de l'Eau et de l'Assainissement (ONEA)
Project features: Water supply feasibility study for the identification of viable alternatives to secure the drinking water supply to Ouagadougou through 2030 and beyond. Funded by the World Bank.
Position held: Technical Specialist
Activities: Geospatial analysis and GIS mapping for the multicriteria selection and conceptual design of reservoir sites at the national level.

Zoti HPP Cascade Technical Due Diligence

Date: 2021-2022
Location, Country: Guria, Georgia
Client: Energy Development Georgia Ltd.
Project features: Technical due diligence of the Zoti HPP 1 (21.3 MW) and Zoti HPP 2 (24.7 MW) run-of-river hydropower plants.
Position held: Technical Specialist
Activities: Site inspection and review of the detailed designs of the civil structures associated with the hydropower plants.

Design Services for Lower Namakhvani Dam

Date: 2021

Location, Country: Racha-Lechkhumi, Georgia
Client: Enmar Construction B.V.
Project features: Detailed design of the 120 m high RCC gravity Lower Namakhvani dam and its appurtenant structures.
Position held: Deputy Project Manager
Activities: Management of the project's technical and administrative aspects, coordination of the multidisciplinary design team, monitoring of project progress, quality management of project deliverables, and provision of technical support to the Client.

Kariba Dam Rehabilitation Project

Date: 2021
Location, Country: Zambia and Zimbabwe
Client: Zambezi River Authority (ZRA)
Project features: Detailed design and construction supervision of rehabilitation works for the spillway and plunge pool of the 128 m high double-curvature arch Kariba dam. Funded by the World Bank.
Position held: Technical Specialist
Activities: Review of the contractor's technical documentation, geospatial analysis, and hydrological assessment of surface runoff into the plunge pool in support of the design of the site dewatering system.

South Lincolnshire Reservoir Concept Design

Date: 2021
Location, Country: Lincolnshire, United Kingdom
Client: Anglian Water
Project features: RAPID Gate 1 concept design for a 50 MCM water supply reservoir in Lincolnshire.
Position held: Design Lead
Activities: Coordination of the multidisciplinary design team and technical leadership of the civil hydraulic design of the concept options, including reservoir earthworks, ancillary structures, and flood storage areas. Geospatial analysis and 3D CAD modelling for the estimation of quantities for the main civil structures.

Ouse Washes Section 10 Works ECC Supervision

Date: 2020-2021
Location, Country: Cambridgeshire and Norfolk, United Kingdom
Client: Environment Agency (EA)
Project features: Construction supervision for the raising of embankments forming part of the 90 MCM Ouse Washes flood storage reservoir.
Position held: Site Inspector
Activities: Inspection and supervision of construction works, monitoring of quality control, and verification of compliance with site health and safety requirements.

New Rochelle Reservoir No. 1 Dam Stability Analysis

Date: 2020
Location, Country: New Rochelle, United States of America
Client: City of New Rochelle
Project features: Feasibility study for the rehabilitation of a reservoir impounded by two earth embankments and a 30 ft high masonry dam.
Position held: Design Engineer
Activities: Static and seismic stability analysis of the masonry dam using CADAM2D. Assessment of the existing condition and evaluation of potential rehabilitation measures to inform the definition and scope of the required remedial works.

Great Yarmouth Flood Defences Epoch 2

Date: 2019-2020
Location, Country: Great Yarmouth, United Kingdom
Client: Environment Agency (EA)
Project features: Construction services for the rehabilitation of the tidal flood defence walls along the Yare and Bure rivers.
Position held: Site Engineer
Activities: Setting-out, levelling, and surveying; review of plans and drawings; coordination of construction activities; procurement of materials, plant, and subcontractors; and oversight of quality control and compliance with site health and safety requirements.

Lluest Wen Spillway CFD Modelling

Date: 2019
Location, Country: Rhondda Cynon Taf, United Kingdom
Client: Dŵr Cymru Welsh Water (DCWW)
Project features: Detailed design of rehabilitation works for the spillway of the 24 m high earth embankment Lluest Wen dam.
Position held: Modelling Engineer
Activities: 3D CAD and computational fluid dynamics (CFD) modelling of the spillway using FLOW-3D. Assessment of its hydraulic performance under the 10,000-year design flood and probable maximum flood conditions in support of the definition of improvement and remedial works.

Dam Break Analysis for the Zambezi River Basin

Date: 2018-2019
Location, Country: Mozambique, Zambia, and Zimbabwe
Client: Zambezi River Authority (ZRA)
Project features: Quantitative risk assessment and dam break study of the potential consequences of breach failures at seven dams in the Zambezi River basin: Kariba, Cahora Bassa, Itezhi-Tezhi, Kafue Gorge Upper, Kafue Gorge Lower, Batoka Gorge, and Mphanda Nkuwa dams. Funded by the World Bank.
Position held: Modelling Engineer
Activities: Dam breach and 2D hydraulic modelling using HEC-RAS, preparation of flood inundation maps, and geospatial spatial analysis.

Xe-Pian Xe-Namnoy Hydropower Project – New Saddle Dam

Date: 2018-2019
Location, Country: Champasak, Laos
Client: SK Engineering & Construction
Project features: Detailed design of a 53 m high RCC dam to restore the design storage capacity of the Xe-Namnoy reservoir following the collapse of Saddle Dam D.
Position held: Modelling Engineer
Activities: 3D computational fluid dynamics (CFD) modelling using FLOW-3D for the analysis and optimisation of the hydraulic design of the dam's mid-level outlet and flip bucket.

Earth Observation Data for Renewable Energy Technologies Study

Date: 2018-2019
Location, Country: Brussels, Belgium
Client: European Commission Directorate-General for Research and Innovation
Project features: Research study undertaken by RINA Consulting and EY to assess research needs associated with the use of Earth Observation data to support the development and deployment of renewable energy technologies.
Position held: External Expert
Activities: Participation in stakeholder validation workshops as a hydropower subject-matter expert.

Beckton Sewage Treatment Plant Inlet Works Upgrade

Date: 2018-2019
Location, Country: London, United Kingdom
Client: Thames Water
Project features: Detailed design for the upgrade and expansion of Beckton Sewage Treatment Works to increase the treatment capacity of the facility.
Position held: Modelling Engineer
Activities: 3D computational fluid dynamics (CFD) modelling using FLOW-3D of the inlet works from the forebay to the tailbay, including the screens and grit channels.

Heathrow Expansion Project

Date: 2018
Location, Country: London, United Kingdom
Client: Heathrow Airport
Project features: Concept design of the river diversions and flood storage areas associated with the proposed third runway expansion of Heathrow Airport.
Position held: Design Engineer

Activities: Hydraulic design of the river diversion channels and flood storage areas associated with the planned third runway.

Houghton Regis Flood Storage Area

Date: 2018-2020
Location, Country: Luton, United Kingdom
Client: Environment Agency (EA)
Project features: Detailed design of the 171,000 m³ Houghton Regis flood storage area.
Position held: Design Engineer
Activities: Detailed civil hydraulic design and structural stability analysis of the reservoir's culvert control structure.

HS2 Enabling Works

Date: 2018
Location, Country: Southam, United Kingdom
Client: High Speed Two (HS2)
Project features: Flood risk assessment for HS2, the high-speed railway between London and Birmingham.
Position held: Modelling Engineer
Activities: Review of the 1D-2D HEC-RAS hydraulic model of the railway viaduct crossing the Itchen River for flood risk assessment purposes.

WCD Reservoir Panel Engineer Projects

Date: 2018-2019
Location, Country: England, United Kingdom
Client: Bristol Water, Canal & River Trust, Environment Agency (EA), Thames Water, and United Utilities (UU)
Project features: Statutory inspections and dam safety risk assessments of large reservoirs in the United Kingdom under the Reservoirs Act 1975.
Position held: Project Manager
Activities: Support to All Reservoirs Panel Engineers (ARPEs) in the administrative and financial management of projects. Monitoring of project costs, assistance with change control, updating of project programmes, and support for invoicing and internal project management procedures.

Lim Hydropower Projects

Date: 2017
Location, Country: Prijepolje, Serbia
Client: Eren Energy Group d.o.o.
Project features: Feasibility study of the Kolovrat, Pranike, and Brodarevo run-of-river hydropower plants on the Lim River.
Position held: Design Engineer
Activities: Hydrological analysis, 1D-2D river hydraulic modelling using HEC-RAS for hydropower potential assessment and flood inundation mapping, and geospatial analysis.

Iren Energia Hydropower Plants in the Orco Valley

Date: 2017
Location, Country: Turin, Italy
Client: Edison S.p.A.
Project features: Feasibility study in support of the potential acquisition of the operating licences for the Ceresole-Rosone HPP (98.6 MW), Rosone-Bardonetto HPP (18 MW), and Bardonetto-Pont HPP (14.6 MW) run-of-river hydropower plants, and the Agnel-Serrù-Villa HPP (40 MW) and Telessio-Eugio-Rosone HPP (82.4 MW) storage hydropower plants in the Orco River basin.
Position held: Design Engineer
Activities: Hydrological analysis for hydropower potential and environmental impact assessment, and geospatial analysis.

Riverbank Protection Works on the Borlezza River

Date: 2017
Location, Country: Cerete, Italy
Client: Idroelettrica Lombarda S.r.l.
Project features: Detailed design of riverbank protection works along a landslide affected reach of the Borlezza River.
Position held: Design Engineer

Activities: Hydrological analysis, 1D river hydraulic modelling using HEC-RAS for flood hazard assessment, and geospatial analysis.

Licence Renewal of Bovegno Hydropower Plant

Date: 2017
Location, Country: Bovegno, Italy
Client: Rothaug Albrecht S.r.l.
Project features: Feasibility study for the renewal of the operating licence for a 0.3 MW run-of-river hydropower plant on the Mella River.
Position held: Design Engineer
Activities: Hydrological analysis for the environmental impact assessment and geospatial analysis.

Campiglia Cervo Hydropower Project

Date: 2017
Location, Country: Campiglia Cervo, Italy
Client: Granda Energie #2 S.r.l.
Project features: Feasibility study of a 1.6 MW run-of-river hydropower plant on the Cervo River.
Position held: Design Engineer
Activities: Civil hydraulic design of the diversion weir, site inspection, geospatial analysis, hydrological analysis, and 1D-2D river hydraulic modelling using HEC-RAS for hydropower potential assessment, environmental impact assessment, and flood hazard assessment.

Južna Morava Hydropower Projects

Date: 2016-2017
Location, Country: Aleksinac, Serbia
Client: Pessina Gestioni S.r.l.
Project features: Feasibility study of the Bobovište, Tešica, and Glogovica run-of-river hydropower plants on the Južna Morava River.
Position held: Design Engineer
Activities: Hydrological analysis, 1D river hydraulic modelling using HEC-RAS for hydropower potential assessment and flood inundation mapping, and geospatial analysis.

Technical Due Diligence and Valuation of Small Hydropower Plants in Italy

Date: 2016-2017
Location, Country: Northern Italy
Client: BKW Italia S.p.A.
Project features: Technical due diligence of run-of-river hydropower plants in the Aosta Valley: Centrale Messuère SHP (0.4 MW) and Rû Gontier SHP (0.2 MW); Piedmont: Roggia Natta SHP (1.5 MW); Lombardy: Centrale Stabina-Caravino SHP (1.2 MW); and Emilia-Romagna: Castellarano SHP (3.3 MW) and Villalunga SHP (2.1 MW).
Position held: Design Engineer
Activities: Site inspections, review of the technical designs and existing condition of the hydropower plants, hydrological analysis for hydropower potential assessment, and geospatial analysis.

Rio Claro 2 Hydropower Project

Date: 2016-2017
Location, Country: Maule, Chile
Client: Electrade S.p.A.
Project features: Detailed design of a run-of-river hydropower plant on the Claro River.
Position held: Design Engineer
Activities: Civil hydraulic design of the intake structure and headrace channel.

Santa Maria Magale Hydropower Project

Date: 2016-2017
Location, Country: Terni, Italy
Client: ERG Hydro S.r.l.
Project features: Feasibility study of a 0.2 MW run-of-river small hydropower plant on the Nera River.
Position held: Design Engineer

Activities: Technical support for the environmental impact assessment.

Mazzunno Hydropower Project

Date: 2016-2017
Location, Country: Colere, Italy
Client: Italgen S.p.A.
Project features: Feasibility study for the rehabilitation of the intake structure of a 6.4 MW run-of-river hydropower plant on the Dezzo River.
Position held: Design Engineer
Activities: Site inspection, civil hydraulic design of the intake structure, hydrological analysis for hydropower potential assessment, and geospatial analysis.

Licence Renewal of Prevalle-Naviglio Hydropower Plant

Date: 2016-2017
Location, Country: Prevalle, Italy
Client: Chi.Na.Co. S.r.l.
Project features: Feasibility study for the renewal of the operating licence for a 1.0 MW run-of-river hydropower plant on the Naviglio di Brescia Canal.
Position held: Design Engineer
Activities: Site inspection, hydrological analyses for hydropower potential assessment, and 1D hydraulic modelling of the headrace channel using HEC-RAS.

Technical Assessment of Gaetano Ganassini Dam

Date: 2016
Location, Country: Quincinetto, Italy
Client: U.E.I. S.p.A.
Project features: Dam safety assessment of a 10.75 m high concrete diversion dam on the Dora Baltea River used for hydropower generation.
Position held: Design Engineer
Activities: Seepage analysis using GMS/SEEP2D, static and seismic stability analysis of the dam, and 1D river hydraulic modelling using HEC-RAS for flood hazard assessment.

Technical Due Diligence and Valuation of Digon Hydropower Plant

Date: 2016
Location, Country: San Nicolò di Comelico, Italy
Client: Sviluppo Energie Alternative S.p.A.
Project features: Technical due diligence of a 0.8 MW run-of-river hydropower plant on the Digon River.
Position held: Design Engineer
Activities: Review of the technical design and existing condition of the hydropower plant, hydrological analysis for hydropower potential assessment, and geospatial analysis.

Bolognano Hydropower Project

Date: 2016
Location, Country: Bolognano, Italy
Client: Compagnia Elettrica Italiana S.r.l.
Project features: Feasibility study of a 0.1 MW run-of-river hydropower plant on the Orta River.
Position held: Design Engineer
Activities: Technical support for the environmental impact assessment.

Technical Due Diligence and Valuation of Palanzano Hydropower Plant

Date: 2016
Location, Country: Palanzano, Italy
Client: Aventron AG
Project features: Technical due diligence of a 0.5 MW run-of-river small hydropower plant on the Cedra River.
Position held: Design Engineer
Activities: Review of the technical design and existing condition of the hydropower plant, hydrological analysis for hydropower potential assessment, and geospatial analysis.

Pretara Hydropower Project

Date: 2016
Location, Country: Isola del Gran Sasso, Italy
Client: Absolute Energy Capital LLP
Project features: Feasibility study of a 0.8 MW run-of-river hydropower plant on the Ruzzo River.
Position held: Design Engineer
Activities: Hydrological analysis for hydropower potential assessment and geospatial analysis.

Les Iles Hydropower Project

Date: 2016
Location, Country: Jovençan, Italy
Client: Atelier Projet Studio Associato
Project features: Technical advisory services for the optimisation of the civil hydraulic design of a run-of-river hydropower plant on the Dora Baltea River.
Position held: Design Engineer
Activities: Civil hydraulic design of the intake and outlet structures.

Marengo Hydropower Project

Date: 2016
Location, Country: Marmirolo, Italy
Client: Marengo Hydro S.r.l.
Project features: Feasibility study of a 0.2 MW run-of-river hydropower plant on the Scaricatore di Pozzolo Canal.
Position held: Design Engineer
Activities: Seepage analysis using GMS/SEEP2D, static and seismic stability analysis of the diversion weir, and 1D hydraulic modelling of the hydropower plant civil structures using HEC-RAS.

Soave Hydropower Project

Date: 2016
Location, Country: Porto Mantovano, Italy
Client: Soave Hydro S.r.l.
Project features: Feasibility study of a 0.7 MW run-of-river hydropower plant on the Scaricatore di Pozzolo Canal.
Position held: Design Engineer
Activities: 1D hydraulic modelling of the hydropower plant civil structures using HEC-RAS.

Santa Caterina Hydropower Project

Date: 2016
Location, Country: Verona, Italy
Client: AGSM Verona S.p.A.
Project features: Feasibility study of a 1.3 MW run-of-river hydropower plant on the Adige River.
Position held: Design Engineer
Activities: Site inspection and 1D hydraulic modelling of the hydropower plant civil structures using HEC-RAS.

Val Danerba Hydropower Project

Date: 2016
Location, Country: Valdaone, Italy
Client: Comune di Valdaone
Project features: Detailed design of a 1.0 MW run-of-river hydropower plant on the Danerba River.
Position held: Design Engineer
Activities: Civil hydraulic design of the hydropower plant structures.

Travata di Gottolengo Hydropower Project

Date: 2016
Location, Country: Gottolengo, Italy
Client: SAIR S.r.l.
Project features: Feasibility study of a 0.1 MW run-of-river hydropower plant on the Gambara River.
Position held: Design Engineer

Activities: Site inspection, seepage analysis using GMS/SEEP2D, and static and seismic stability analysis of the diversion weir.

Reconstruction of the Barrage on the Sauro River in Agro di Aliano

Date: 2015
Location, Country: Aliano, Italy
Client: Pro.Cos.Ma S.c.a.r.l.; Pistorello S.p.A.
Project features: Feasibility study for the reconstruction of a 12 m high concrete dam on the Sauro River for water supply and irrigation.
Position held: Modelling Engineer
Activities: CAD drawings preparation and 2D river hydraulic and sediment transport modelling using InfoWorks ICM.